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College of Information Technology Education

CorePOS: Unified Sales and Membership Web-Platform for Gym

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By:

Jerez, Janren

Punsalan, Prince Sydrick

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The findings of the platform evaluation and user testing include: 32

- All-In-One Operation: The system allows users to manage membership, conduct check-ins and conduct sales operations in one tool. There is no need for staff to open other applications or use paper logbooks. 32
- Automatic Membership Expiration Dates Checking: The system checks membership expiration date, so there is no possibility of allowing access of non-paying or expired members. 32

- Faster Check-Ins: The system allows fast check-in of members via barcode scanning which takes less than a few seconds. This means no lines on the front desk during gym peak hours. 18
- Real-Time Sales and Inventory Management: With every sale, the inventory is updated and sales are recorded in the system. The system allows for accurate tracking of revenue and prevents losses and errors due to manual sales accounting. 32
- Role-Based Access Control: Only the users who need to see certain information will be able to do that because of this system. Users can process sales and check-ins, but only administrators can view financial statements. 32
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There is much value in using the Unified Sales and Membership Web Platform for both the strategic and technical aspects of gym management. Gym owners will benefit greatly from this application since it will end up ensuring that no money is lost due to inefficient operations or because of an untidy record-keeping system. This is made possible by integrating sales transactions, membership renewal, and inventory information into one centralized database. With real-time sales and attendance records, gym managers can easily make important decisions regarding business matters and also cut down on licensing costs for software. On the other hand, the staff at the reception desk will benefit from the fact that most of their work will be automated through the use of barcodes and expiration checks. 34

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CHAPTER 1

INTRODUCTION

1.1 Background of the Study

The use of web-based systems has become common in many businesses today. These systems help improve operations, manage data, and make processes faster and more efficient. In the fitness industry, gyms need proper systems to handle memberships, sales, and customer records.

Project E Fitness is a gym that offers memberships and fitness-related products. Currently, the gym manages its records manually or using separate systems. This setup makes it difficult to track memberships, monitor sales, and verify members during peak hours.

Because of this, the staff experience delays in checking members, and there is a chance of missing expired memberships which can affect revenue. Reports are also difficult to generate since the data is not centralized.

To solve this problem, the researchers propose a system called CorePOS. This is a web-based system that integrates membership management, sales transactions, and check-in processes into one platform. The system will help improve efficiency and provide better monitoring of the gym operations.

Technical Background of the Study

The contemporary working process of gyms tends to use database-based architecture for minimizing the amount of friction during its work and combining different business processes into one administrative channel.

The design of the Unified Sales and Membership Web Platform for a Gym is based on a multi-layer RDBMS approach and uses structured query processing and server-side processing for ensuring transaction safety, data consistency, and relational integrity of all entities used in the database.

Thanks to centralizing all the operations of the gym, such as registration of users, RBAC, POS inventory reduction, subscription expiration scheduling, and attendance tracking using bar codes, the backend system allows reducing the load on the client side and preventing any synchronization issues that appear when using old-fashioned systems of record-keeping.

1.2 Statement of the Problem

This study aims to solve the following problems:

These problems affect the daily operations of the gym because staff need to handle membership records, sales, and member verification using separate or manual processes. As a result, some tasks may take more time and records may not always be updated or checked immediately.

1. The current system is manual and time-consuming:
   This problem affects the efficiency of the daily operations of Project E Fitness because staff members need to spend additional time handling records and completing routine tasks manually. Manual processes may also make it more difficult to organize information and

complete transactions consistently. The proposed CorePOS platform addresses this concern by providing a centralized web-based system for the gym's main operational activities.

2. Membership and sales records are not centralized:

   The lack of a centralized record system can make it difficult for authorized users to access and manage membership and sales information in an organized manner. Keeping these records in separate or disconnected forms may also require additional effort when checking or reviewing information. CorePOS is intended to bring these important records together in one platform to make information easier to manage and retrieve.

3. Staff have difficulty verifying membership status:

   Verifying membership status is an important part of the gym's daily operations because staff need to determine whether a member has an active membership before providing access to the gym. When this process is handled manually, checking the necessary information may take more time, especially during busy periods. The proposed system provides a more organized way of checking membership information and status.

4. There is no real-time reporting system:

   The absence of a centralized reporting function makes it more difficult for the gym to monitor its current operational information. Organized reports can help authorized users review membership, sales, and other available records without depending entirely on manual compilation. CorePOS is designed to provide basic reports and dashboard information that can support the monitoring of the gym's business activities.

5. Records are prone to errors and data loss:

   Manual handling of records can increase the possibility of incorrect entries, incomplete information, or misplaced records. These issues may affect the accuracy and reliability of information used in daily gym operations. By storing important records in a centralized system, CorePOS aims to provide a more organized method of maintaining membership, sales, and related operational information.

1.3 Objectives of the Study

The main objective of this study is to design, develop, and implement a Unified Sales and Membership Web Platform for a Gym to streamline front-desk operations, automate membership lifecycles, and centralize transaction tracking.

1. Designing and Development of System Modules: Develop an integrated web interface with automated membership management for registrations and renewals, POS (Point of Sale) for managing sales and inventories, and check-in system for attendance tracking . This objective focuses on developing the main system modules needed to support the daily operations of Project E Fitness. The modules are intended to work together through one web-based platform so that membership registration and renewal, sales and inventory activities, and member check-in can be handled in an organized manner. Integrating these functions helps the proposed system address the operational concerns identified in the study.

2. Implement dynamic user roles and permissions tailored for Administrators, Front-Desk Staff, and Gym Members to ensure operational security and data privacy. This objective focuses on organizing access to the system according to the responsibilities of each type of user. Different users may require different functions and information when using the platform, so assigning appropriate roles and permissions helps keep system access organized. This also supports the protection of operational information by limiting access to authorized users based on their assigned responsibilities.

3. Assess the developed platform in terms of processing speed, data accuracy, operational efficiency, and user satisfaction through structured User Acceptance Testing (UAT) with gym staff and administrators. This objective focuses on evaluating whether the developed platform performs according to its intended purpose. The assessment considers important aspects such as processing speed, accuracy of information, efficiency of operations, and the experience of the intended users. User Acceptance Testing provides an opportunity for gym staff and administrators to evaluate the system based on the functions they are expected to use.

1.3.1 General Objective

The general objective of this study is to develop a web-based system that integrates membership, sales, and check-in processes for Gym.

This objective focuses on bringing the main gym operations into one platform so that staff can manage records and transactions more easily and consistently.

1.3.2 Specific Objectives

Specifically, the study aims to:

The specific objectives describe the main functions and improvements that the researchers want the system to achieve for Project E Fitness.

1. To design a centralized system for managing membership and sales. This objective focuses on providing a single platform where authorized users can manage important membership and sales information. Centralizing these functions can help reduce the need to maintain separate records and can make daily transactions easier to organize. It directly supports the main purpose of CorePOS as a unified sales and membership web-platform for the gym.
2. To automate the check-in and membership verification process. This objective focuses on making the member check-in process more organized and efficient. The system is intended to assist staff in verifying membership information while recording member check-ins through the available check-in function. This can help reduce the time spent on manual verification during regular gym operations.
3. To record and track sales transactions. This objective focuses on maintaining organized records of sales transactions made through the gym. Recording transactions in the system allows authorized users to keep sales information in a centralized location for easier monitoring and review. This also supports the need of Project E Fitness to organize its sales-related records.
4. To generate simple reports for monitoring business performance. This objective focuses on providing basic reports that can help authorized users review available business information. The reports are intended to organize relevant records such as sales and membership information so that the gym owner and authorized staff can monitor daily operations more conveniently. The objective is limited to basic reporting based on the documented scope of CorePOS.
5. To improve efficiency and reduce manual work. This objective focuses on reducing repetitive manual tasks involved in managing the gym's daily operations. By bringing membership, sales, check-in, and basic reporting functions into one web-based platform, CorePOS can provide a more organized way of completing common operational tasks. This supports the overall goal of improving the efficiency of Project E Fitness.

1.4 Scope and Delimitation

The **Unified Sales and Membership Web-Platform for Gym f**ocuses on centralizing and automating core fitness center operations through a responsive web application. The system will include:

The scope covers the main activities needed for membership handling, sales recording, and member check-in. These features are included to help reduce manual work and make the daily operations of the gym more organized.

- Membership management
  - Membership management is included to provide an organized way of maintaining member information and membership records. This function supports activities related to managing memberships and helps authorized gym staff access the information needed for regular membership-related transactions.
- Sales recording
  - Sales recording is included to allow authorized users to record and maintain information about transactions made within the gym. This function supports the organization of sales records and helps provide a centralized record of the gym's sales activities.
- QR-based check-in
  - QR-based check-in is included to support a more organized method of recording member attendance and checking membership information. This function is intended to assist gym staff during the check-in process and reduce the need for completely manual attendance recording.
- Basic reports and dashboard
  - Basic reports and dashboard functions are included to provide authorized users with organized information about the gym's available operational records. These functions can help the gym owner and staff monitor basic sales and membership information and support their review of daily business activities.

The study will not include:

Some features are outside the current scope of the study because the project focuses on the core operations needed by the client.

- Online payment system
  - The online payment system is not included because the current study focuses on the core sales and membership operations of the gym. Payment-related transactions within the system are limited to recording and managing the information needed for the gym's regular operations. This keeps the development focused on the requirements identified for Project E Fitness.
- Mobile application
  - A dedicated mobile application is not included in the current scope because CorePOS is developed as a web-based platform. The system is intended to be accessed through supported web browsers, allowing authorized users to use the platform without requiring a separate native mobile application. This limitation keeps the development focused on the main web-based functions of the system.
- Advanced analytics or AI features
  - Advanced analytics and artificial intelligence features are also outside the current scope of the study. CorePOS focuses on basic reporting and organized business information rather than predictive or AI-based analysis. This allows the researchers to concentrate on the essential membership, sales, check-in, and reporting functions required by the client.

The system is intended for:

The system is mainly intended for the people who handle and use the gym's daily operations. Their access will depend on their role in the system.

- Gym staff
  - Gym staff are intended to use the system for the operational tasks assigned to them. These tasks may include managing member information, checking membership status, recording sales transactions, and processing member check-ins. Their access to the system will depend on the permissions assigned to their role.
- Gym owner
  - The gym owner is intended to use the system to monitor important information related to the gym's operations. This may include reviewing membership records, sales information, and basic reports provided by the platform. The owner's access will be based on the appropriate role and permissions within the system.

The platform covers user authentication and role-based access control for authorized users, membership subscription management, point-of sale (POS) transactions for gym merchandise and services, QR-based check-in, and basic sales and membership reporting.

The system is designed as a web-based platform that can be accessed through supported desktop and mobile browsers. It does not include a dedicated native mobile application, online payment gateway, or advanced analytics and AI features. Inventory tracking is limited to gym managed merchandise and products.

1.5 Significance of the Study

Administrators (Gym Owner/Management)
The system will help the management of Project E Fitness to easily monitor memberships, sales, and reports in one platform. It will reduce manual work and improve decision-making through organized data.

- Gym Managers and Owners: This platform acts as an important decision-making tool that brings all the operations of the business on a single interface. Managers can be updated about sales trends, expiry dates of memberships, performance of the staff members, and inventory turnover rate. This not only saves time and efforts but also ensures data analysis to maximize the income generation of the business.
- Gym Staff and Front-desk personnel: The system helps in managing the day-to-day activities of the gym by automatically updating the memberships and performing financial transactions such as validations and renewals. Staff members can check-in customers using digital QR Codes.

- Gym Members: The subscribers will have the ease of using the digital platform for account management, renewing the subscription, buying the items from the store, and scheduling the training sessions according to their convenience.
- Future Researchers and Developers: This paper will act as an important reference for researchers and developers who aim to develop future versions of such platforms for fitness management by incorporating features like secure payment gateways, access control mechanism, and automated inventory system through a single web interface.

1.6 Definition of Terms

POS (Point-of-Sale): The web platform includes an integrated POS module to instantly record sales transactions for gym merchandise and membership plan purchases.

Membership: System administrators can efficiently manage new member registrations and track subscription plan durations in real time.

Check-in: Front-desk staff use a streamlined check-in process to verify member entry status in seconds through QR-based check-in.

Web-Based System: The platform operates as a secure web-based system, allowing staff and administrators to access operational tools from any device using a web browser.

Database: All sales records, member details, and inventory counts are stored securely within a central database to guarantee data integrity.

Dashboard: The administrator dashboard summarizes basic sales, membership, and system information to help management monitor business performance.

For example, a product record can be connected to a sale through the product ID. A member can be connected to a check-in through the member ID. These relationships help the system maintain consistent records across different modules.

Using identifiers to connect records reduces unnecessary duplication. It also makes updates easier because related information can be retrieved from its source.

The tables are connected because the business processes are connected. A member can have check-in records. A product can appear in sales transactions. A user account has a role that controls access.

Relationships Between Core Tables

This is a simple form of business analytics. It does not require a machine-learning model. The important part is that daily business data is stored consistently so the system can summarize it.

The sale ID identifies the transaction, while the product ID connects the record to the product. Quantity and price provide the basic values needed for sales analysis.

The current Analytics table includes an ID, sale ID, product ID, quantity, and price. These fields can support summaries of sales activity.

Analytics Table

Structured records allow the system to retrieve only the information needed for a particular task. This reduces repeated encoding and gives reports a more consistent data source.

These fields support membership identification and verification. The status field is important during check-in because it provides a current state that can be checked before an entry is recorded.

The current transaction-related structure contains information such as QR ID, name, contact, address, membership type, start date, expiry date, status, and created date.

Transaction Details Table

The stock value is especially important when a product is sold. If the system updates stock after a completed transaction, management can monitor products that are becoming low. This connects the inventory function directly to the POS process.

The product ID is the main identifier. The name and price describe the product, while stock represents the quantity available. The image URL can help display a product in the interface.

The Products table contains the information needed for merchandise management. The current design includes ID, product name, price, stock, and image URL.

Products Table

The table should use validation so that a scan is connected to a valid member. Incorrect identifiers could create incorrect attendance information. Keeping the check-in data structured also makes it easier to review attendance later.

The member ID connects the attendance record to the correct member. Scan time records when the member was checked in. Multiple visits can therefore be stored as separate attendance events.

The Customers table is connected to member information and check-in activity. The current design includes an ID, member ID, and scan time

Customers Table

Sensitive account information should be protected. Passwords should not be stored as plain text in a real deployment. They should be protected using appropriate password-hashing practices. Access to user records should also be limited to authorized administrative functions.

The ID identifies the account. The username is used during login, while the role determines the account's permissions. The created date records when the account was created.

The Users table stores account and access information. The current design includes ID, username, password, role, recovery question, recovery answer, and created date.

Users Table

This structure helps reduce repeated information and makes reporting easier. Instead of storing every detail in one large record, the system stores information in related tables and retrieves what is needed. A well-organized database is therefore a major part of the project.

Each table has a specific purpose. Related records can be connected using identifiers. For example, a transaction can refer to a product, while a check-in can refer to a member.

The database is the central storage area of CorePOS. The current document identifies tables for users, customers, products, transaction details, and analytics.

Database Design Overview

RBAC supports security because users are not automatically given access to every part of the database. It also makes account management easier. When a new staff account is created, the administrator can assign the correct role and the system can apply the appropriate permissions.

Administrators need broader access because they manage the system and review business information. Staff need access to daily functions such as check-in and sales. Members need limited access to their own account information.

Role-Based Access Control, or RBAC, separates system permissions according to user roles. CorePOS identifies administrators, staff, and members.

Role-Based Access Control

Analytics therefore begins with accurate data entry. If the source records are incomplete or wrong, the report will also be affected. CorePOS first creates an organized data foundation and then uses that information for monitoring.

Sales can be grouped by date or product. Memberships can be grouped into active and inactive records. Check-ins can be reviewed by day or period. These simple summaries already provide useful information because the owner can compare actual records instead of relying only on memory.

Business analytics in CorePOS means using recorded business information to support decisions. The project is not described as an advanced AI system. Its current scope focuses on organized data, basic reports, and dashboards.

Analytics in CorePOS

The dashboard can present selected information in a simple visual form. The purpose is not to make the system complicated. It is to give the owner or administrator a quicker view of the business condition.

Examples include sales summaries, membership status, and attendance information. A report is useful when it answers a business question. Management may want to know which products sell more often, how many members are active, or how sales change over time.

The reporting function converts stored transactions into information that management can review. The current project describes basic reports and a dashboard for business monitoring.

Reports and Dashboard

Inventory is connected to the POS because a sale is not only a financial event. It also represents a change in the quantity available. Keeping these records together gives the owner a better view of products sold and products remaining.

When a product is sold, the transaction identifies the product and quantity. The system can use this information to reflect the change in stock. This is useful for gym merchandise such as supplements, apparel, and other products.

CorePOS includes product and transaction information that can support basic inventory monitoring. The current Products table includes an ID, name, price, stock, and image URL.

Inventory-Related Processing

Check-in records can also support business analysis. Management can review how often the gym is used and which periods are more active. The project keeps this within basic reporting rather than claiming advanced artificial intelligence.

The normal process is simple: the member identifier is scanned or entered, the system retrieves the membership record, and the membership status is checked. If the member is active, the visit can be recorded. If the membership is inactive or expired, the staff can see the status before continuing.

The check-in function is intended to make attendance verification faster. The existing document describes QR or barcode-based checking.

Check-In Module

The main benefit is consistency. Staff do not need to write the sale in a logbook and later transfer it into another report. The transaction becomes available in the database as soon as it is recorded.

A basic transaction starts when staff select a product or service and enter the quantity. The system checks the information and saves the transaction. If inventory is being monitored, the sale can also be used to update the available quantity.

The POS component records sales transactions. The current database design includes products, prices, quantities, and transaction details. This allows the gym to record merchandise or service sales in a structured way.

Sales and POS Module

The module therefore helps the gym with both daily operations and later analysis.

Centralized membership records can also support monitoring of active and inactive members. If a member renews, the record can be updated instead of creating another disconnected record. These records can later support reports about membership activity.

The system can store basic member details, membership type, start date, expiration date, and status. The expiration date is important because it gives the system a basis for deciding whether a membership is still active.

Membership management is one of the main parts of CorePOS. It provides a structured place for member information and membership status.

CHAPTER II

REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Related Literature

1. Software Engineering

According to Pressman and Maxim (2019), software engineering is a systematic approach to developing software to ensure quality, reliability, and maintainability. Following a structured software development process helps organizations create systems that solve real business problems.

Relevance to the Study:

This supports the development of CorePOS because the proposed system follows a structured approach in developing a web based sales and membership management system.

Reference:

Pressman, R. S., & Maxim, B. R. (2019). Software Engineering: A Practitioner's Approach (9th ed.). McGraw-Hill.

2. Management Information Systems

According to Laudon and Laudon (2020), management information systems help organizations collect, process, store, and distribute information to support decision-making and improve business operations.

Relevance to the Study:

CorePOS provides organized information about memberships, sales, and reports to help the gym owner make better decisions.

Reference:

Laudon, K. C., & Laudon, J. P. (2020). Management Information Systems: Managing the Digital Firm (16th ed.). Pearson.

3. Database Management Systems

According to Connolly and Begg (2015), database management systems reduce data redundancy, improve data accuracy, and make information easier to retrieve.

Relevance to the Study:

CorePOS uses a centralized database to securely store member records, transactions, and product information.

Reference:

Connolly, T., & Begg, C. (2015). Database Systems: A Practical Approach to Design, Implementation, and Management (6th ed.). Pearson.

2.2 Related Studies

1. Gym Management System

Chokhandre, Raut, Chhatre, and Tarekar (2025) developed a Gym Management System that automates membership management, payment processing, attendance monitoring, and reporting. Their study found that replacing manual processes with a digital system improved efficiency and provided useful analytics for gym owners.

Relevance to the Study:

Like their study, CorePOS automates gym operations, but it specifically focuses on integrating sales, memberships, and check-in into one platform.

2. Q-Fitness Gym System

Hussain, Wei Yi, Halimi, and Syed Abidin (2025) Accordng to Hussan developed the Q-Fitness Gym System, a web-based platform that centralizes member registration, scheduling, staff management, and payment processing. Their study showed that integrating gym operations into one web application improves operational efficiency and member engagement [1].

Relevance to the Study:

This supports CorePOS because both systems aim to centralize gym operations and improve service quality.

3. Fit Drive Management System

The Fit Drive Management System (2026) demonstrated that automating member records, reports, and revenue monitoring reduced manual record-keeping errors and improved data organization.

Relevance to the Study:

CorePOS also automates member management and reporting to improve operational lefficiency.

2.3 Synthesis of the Reviewed Literature and Studies

The reviewed literature explains the importance of software engineering, information systems, and database management in developing reliable business systems. The reviewed studies show that web-based gym management systems improve efficiency by automating memberships, sales, attendance, and reporting.

Based on these findings, the proposed CorePOS: Unified Sales and Membership Web-Platform combines these concepts into one centralized system that helps Project E Fitness manage memberships, sales, and check-in more efficiently while providing reports that support better decision-making.

REFERENCES

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https://www.ijsrtjournal.com/assetsbackoffice/uploads/article/Gym%2BManagement%2BSystem.pdf?

[5] Q-Fitness Gym System (2025) https://rsisinternational.org/journals/ijriss/view/q-fitness-gym-system-an-integrated-web-application-for-smart-gymoperations-and-member-engagement?

The existing capstone reports that the main functions were tested and evaluated. The final supporting evidence should include the actual test records used by the group.

For example, staff can log in, locate a member, check membership status, record a check-in, process a sale, and review the stored transaction. This type of testing can reveal problems that do not appear when modules are tested separately.

System testing examines the platform as a complete application. The tester can follow a normal business process from login to transaction.

System Testing

The team should record the actual result for each integration test. Keeping the evidence makes it easier to explain the system during a defense because the group can show that the modules were tested as connected processes rather than only as separate pages.

A member record should support the check-in process. A product record should support a sales transaction. A sale should be connected to product information. A user account should receive the correct permissions based on role.

Integration testing checks whether modules work together. This is important because the main purpose of CorePOS is integration.

Integration Testing

Example: when valid member information is entered, the expected result is that the member record is saved. When an active member is checked in, the expected result is that the attendance event is recorded. These cases should be supported by the team's actual testing evidence.

A unit test should identify the input, the action, and the expected result. The actual result should come from the team's test record.

Unit testing checks individual functions before they are treated as part of the complete system. For CorePOS, the team can document tests for login validation, member registration, membership status, product records, sales entry, and check-in.

Unit Testing Documentation

For CorePOS, important functions include login, membership records, check-in, sales, products, reports, and access permissions. The existing document evaluates the platform using functionality, usability, reliability, efficiency, and security.

Unit testing checks individual functions or modules. System testing checks whether the complete application works as a connected system. User acceptance testing allows intended users to assess whether the system supports their actual tasks.

The project identifies unit testing, system testing, and user acceptance testing. Each level has a different purpose.

Testing Overview

The important point is how the tools work together. The team needs an editor, a server environment, a database, a browser, and version control to support development.

Visual Studio Code can be used for source-code editing. PHP supports server-side development. MySQL provides relational database storage. XAMPP can provide a local development environment. Google Chrome can be used for browser testing, while Git supports version control.

The current document identifies Visual Studio Code, MySQL, PHP, Google Chrome, Git, and XAMPP among the development tools.

Tools and Technologies

Keeping business logic on the server helps apply the same rules to requests from different users. The database remains the central source of stored information.

For a sales transaction, the browser sends the product and quantity. The server processes the transaction and stores the information in the database.

A normal interaction starts in the browser. For example, staff may select check-in and submit a member identifier. The browser sends the request to the application, which checks the member record and returns the membership status.

Browser and Server Interaction

This architecture supports centralized management because users do not need separate copies of business records. Updates are stored in one database and can be accessed by authorized users.

The application layer contains business logic and system functions. The data layer contains the stored records. When a user sends a request, the application validates it, performs the required operation, and returns the result.

CorePOS uses a client-server style architecture. Users interact with the system through a web browser, the application processes requests, and the database stores records.

System Architecture

Accessibility is connected to usability. A system can work technically but still be difficult to use if the interface is confusing. Keeping the design simple supports both staff and members.

Text should remain readable, buttons should be easy to select, and important functions should remain visible when the screen size changes. Forms should use clear labels.

The existing project describes CorePOS as a web-based platform that can be accessed through desktop and mobile browsers. Responsive design is therefore important.

Responsive and Accessible Use

A usable system does not require advanced technical knowledge for normal operations. Short training and a simple user guide can help staff become comfortable with CorePOS.

Membership verification, check-in, and sales should not require unnecessary navigation. Clear labels and consistent buttons can reduce confusion. Error messages should explain what needs to be corrected.

Usability matters because front-desk staff may need to process many tasks during busy hours. Common actions should be easy to find.

Usability Considerations

Recovery should also be documented. A backup is useful only if the team knows how to restore it. The administrator or technical team should keep a simple recovery guide and protect backup files from unauthorized access.

A practical implementation should have a backup procedure. Backups can be made regularly and stored separately from the main database. The schedule should match how frequently the business changes its records.

A centralized database improves organization but also makes the database important to the business. If the database is lost or damaged, several operations can be affected.

Backup and Data Recovery

Security should be considered during development. The system should protect login information and use appropriate practices when communicating with the database. Even a small business system should handle user records responsibly.

Authentication checks whether the user has a valid account. Authorization controls what the user can do after logging in. Sensitive information should not be available to users who do not need it.

CorePOS stores member and transaction information, so security is important. The existing project includes authentication and role-based access control.

Data Security

Validation should happen when data is entered rather than waiting until a report reveals an error. The purpose is not to make the system difficult. It is to prevent avoidable mistakes before they become part of the stored data.

Membership dates should follow a logical order. Quantities should not be negative. Prices should use the correct numeric format. A sales transaction should contain a valid product and quantity.

Data validation is important because reports depend on the quality of the records entered. Required fields should be checked before saving a record.

CHAPTER III:

METHODOLOGY

3.1 Research Design

This study uses a developmental research design because it focuses on creating a system to solve a problem.

3.2 System Development Methodology

The study utilized the Agile Software Development Methodology to design the iterative and interactive software development of CorePOS: Unified Sales and Membership Web-Platform. The methodology was applied to ensure continuous improvement and flexibility throughout the development cycle of the new application by frequently incorporating stakeholder feedback into the development process. In this case, the development team worked with Project E Fitness to improve the proposed features of the software periodically.

Requirements Gathering

During the requirements gathering stage, project E Fitness stakeholders were interviewed, processes were observed, and documents were reviewed by the researchers. They defined the key requirements during this period. The main components were front-desk POS sales logging, membership registration with automated expiration checking, QR check-ins, and administrative reporting dashboards. It is critical to define the functional requirements at this stage. The software will address the gym’s issues of manual logging and insufficient security, including the application of Role-Based Access Control.

Planning

In the planning stage, the project scope was divided into specific elements, which would be achieved in structured agile sprints. In this stage, the team also created a plan regarding the project's development timeline, task distribution, and technical aspects. For instance, the group decided to use PHP and MySQL programming languages and electronic software for desktop application. In addition, the project's cloud-based database management system was designed using Aiven. Overall, the creation of planning sprints helped the team establish technical milestones, address the criticality of features, such as scanners, and prepare for the desktop application and peripherals integration.

Design

The design part of the action plan covered architectural project management, relational database creation, and interface mock-up. Entity-relationship diagrams (ERD) and relational models were developed in MySQL to store information about accounts, members, inventory, transactions, and check-ins. The user interface (UI) and user experience (UX) were designed for the Electron desktop application to ensure that it would have a clean and intuitive design, modern POS displays, convenient scanning statuses, and simple navigation.

Development

During the development stage, some of the system's components were programmed and integrated into the system in accordance with the design specifications. HTML, CSS, and JavaScript were implemented for the front end, while PHP and Visual Studio Code were used for the back end, which involved the coding of server-side procedures, session management, CRUD (Create, Read, Update, Delete) operations, and database queries. The software's application programming interface (API) was linked to a MySQL database, which was hosted on an Aiven server. Finally, the web platform was encapsulated in Electron to form a desktop application for Project E Fitness workstations.

Testing

The testing stage included extensive operational testing of all features, including scanning QR codes, inventory management systems, and user permission settings. The development team performed a series of unit, integration, and functional tests before the official launch of the software. Test data simulation allowed for checking if a QR scanner prompts an immediate check-in of a member, a POS lowers the inventory after a purchase, or the accounting reports are inaccessible to staff. The discovered bugs, database errors, and interface issues were resolved in the next iteration.

Evaluation and Improvement

In the concluding phase of assessment and enhancement, the CorePOS system was implemented in a controlled test environment at Project E Fitness to conduct User Acceptance Testing (UAT). Feedback was collected from administrators, front-desk staff, and gym members through a 5-point Likert scale to gauge the extent to which the software met the criteria of Functionality, Usability, Reliability, Efficiency, and Security. Operational feedback elicited during the evaluation process informed final tweaks to the system such as optimizing the platform's responsiveness during peak usage periods and fine-tuning the automated reporting system. Consequently, the final software solution was able to fully satisfy the tripartite stakeholders' requirements.

3.3 System Architecture

The system uses a client-server setup.

- Users access the system through a browser
- The server processes the data
- Data is stored in a database User Flow:

1. Staff logs in
2. Member is registered or checked
3. QR code is scanned
4. System verifies membership
5. Sales are recorded
6. Reports are generated

3.4 Tools and Technologies Used

HTML, CSS, JavaScript

Hypertext Markup Language (HTML), Cascading Style Sheets (CSS), and JavaScript form the foundational front-end technologies of the CorePOS platform. HTML provides the structural markup for all user interface pages, including the point-of-sale layout, member registration forms, attendance scanning interface, and administrative dashboards. CSS styles these elements, ensuring a responsive, intuitive, and visually consistent interface tailored for fast-paced front-desk operations at Project E Fitness. JavaScript handles client-side interactivity and asynchronous tasks, enabling real-time input processing during QR code scanner verification, dynamic calculation of transaction totals, and UI updates without requiring page reloads.

PHP (hypertext Preprocessor)

PHP serves as the core server-side scripting language that powers CorePOS's application logic and backend processes. It manages user session state, handles authentication and access control between administrative management and front-desk staff roles, and processes incoming requests from client-side forms. PHP scripts process server-side workflows, including CRUD (Create, Read, Update, Delete) operations for member profiles and inventory items, automated check-in logic, sales processing, and the execution of relational database queries.

MySQL

MySQL is the Relational Database Management System (RDBMS) utilized to store, manage, and retrieve structured operational data for CorePOS. It maintains organized relational tables for user credentials, member records, subscription status, product inventory, transaction history, and attendance logs. Through optimized queries and primary/foreign key relationships, MySQL ensures data integrity, executes dynamic reporting aggregations for daily sales summaries, and enforces system rules such as automated membership expiration verification during scanner check-in.

Visual Studio Code

Visual Studio Code (VS Code) was used as the main Integrated Development Environment (IDE) for the architecture, coding, and debugging of CorePOS. The IDE's extension ecosystem, built-in terminal, syntax highlighting, and code-formatting tools helped with modular web development for PHP, JavaScript, CSS, and HTML files. It also allowed for efficient project organization, version tracking, and quick debugging for all of the administrative, staff, and core database components.

Electron (app)

Electron is an open-source framework that makes it easy to turn the CorePOS web application into a standalone desktop app. It works by bundling both the user interface and the PHP backend together inside Chromium and Node.js, so gym staff can open CorePOS right from their desktop, just like any other software. This means they don’t need to rely on web browsers, they get a dedicated workspace, and it’s simpler to connect devices like USB barcode or QR scanners.

Aiven (hosting online)

Aiven is a managed cloud platform that handles hosting and managing the CorePOS MySQL database online. By using Aiven, CorePOS gets secure cloud access, automatic backups, high reliability, and safe remote connections. This setup makes sure that things like sales, membership changes, and attendance logs—collected on the Electron desktop app—are instantly synced to a secure central cloud database, so administrators can easily monitor everything.

3.5 Data Gathering Techniques

Interview with gym owner

The owner of the gym and important members of Project E Fitness's administrative team participated in semi-structured interviews to gather qualitative information about high-level business procedures, operational problems, and strategic needs. Through these conversations, the researchers developed fundamental business guidelines, including membership subscription tiers, policies for enforcing expiration, product pricing structures, and the division of privileges between front-desk and administrative staff positions. The functional design of CorePOS, particularly the logic for automated membership verification, role-based access control (RBAC) privileges, and the

precise metrics required for daily sales and attendance reporting dashboards, was directly influenced by the insights obtained from the interview.

Observation of current process

The initial step in the research involved observing the functioning of the front desk of Project E Fitness to gain insight into the current operating procedures. In particular, the researchers watched how the staff member responsible for the desk performed their work during the slow and busy times of the day. As a result, it was found that they double-check members' personal information in paper logbooks, record sales of retail merchandise manually, and verify subscription details line by line. Consequently, the UI/UX element of CorePOS was designed to resolve the issues discovered during the observations, such as the long entry queue and potential recording errors during peak hours. In particular, the tool was given a quick POS interface and a QR code scanner to reduce the time spent on a single transaction.

Review of existing record

Document analysis was performed on the existing physical and digital records used by Project E Fitness, including manual logbooks, , membership application forms, inventory logsheets, and legacy spreadsheets. By studying these artifacts, the researchers were able to determine the mandatory data fields, underlying transaction workflows, and historical reporting structures. Structural data obtained from these existing records were then used to create the relational database design for CorePOS, which ensures that the MySQL tables (Users, Customers, Products, Transactions, and Attendance) accurately reflect all operational fields required for unified sales tracking, inventory management, and membership lifecycle auditing

3.6 Testing Procedures

- Unit Testing
- System testing
- User Acceptance Testing

3.7 Evaluation Criteria

The system will be evaluated based on:

- Functionality
- Usability
- Reliability
- Efficiency
- Security

CHAPTER IV

SYSTEM DEVELOPMENT, IMPLEMENTATION, TESTING, EVALUATION, RESULTS AND DISCUSSION

The dashboard should therefore focus on information that is actually useful to the client rather than displaying too many unrelated values.

The report does not automatically make the decision. The owner still needs to interpret the information using business knowledge. CorePOS provides organized data so decisions can be based on a clearer view of the operation.

A report should be connected to a business question. A sales report can help management see revenue activity. A product report can help identify items that need attention. A membership report can show active and inactive members. An attendance report can show periods of higher gym activity.

Interpreting Business Reports

This shows where business analytics fits. Analytics starts with collecting useful information correctly and then turning the information into summaries that management can understand.

For example, the business may want to know which products sell more often. This becomes a data requirement involving product IDs, quantities, prices, and transaction dates. Another question may involve membership activity, which requires membership status and check-in records.

CorePOS is also a business analysis project, not only a programming project. The team begins with a business problem and translates that problem into information requirements.

Business Analytics Relevance

The research gap is therefore practical. Similar systems exist, but CorePOS is designed around the workflow and requirements of the selected client. The project applies established information-system ideas to a specific gym environment.

Some systems include mobile applications, online payments, or predictive analytics. These are outside the main current scope of CorePOS. The project focuses on the core operations identified during the client analysis.

The reviewed studies provide examples of gym management, QR attendance, POS, and predictive analytics. However, the exact combination required by Project E Fitness is different.

Research Gap

CorePOS combines the functions most relevant to Project E Fitness: membership management, sales transactions, check-in, and basic reports. The studies support the general approach but do not mean CorePOS is identical to them.

Gym-related studies show the value of web systems for membership, attendance, payment, and administration. POS studies show the value of computerized transaction and stock monitoring.

The reviewed studies show a common pattern. Businesses that depend heavily on manual records can experience delays, errors, and difficulty producing organized information.

Synthesis of Reviewed Studies

The study is therefore useful for the POS part of the project, while gym-specific studies support the membership and attendance parts.

This supports the use of POS technology for reducing manual work. CorePOS has similar sales-recording needs but combines them with membership and check-in functions.

Magbanua, Gealon, Lucero, Mocero, and Villa (2007) developed the Mang Inasal Incorporated Automated Point-of-Sale System. The study aimed to process transactions accurately and quickly, reduce human error, simplify sales computation, print receipts and reports, and maintain customer information.

Local Study: Magbanua and Colleagues

Although the business was a restaurant, the study is related to CorePOS because both involve sales transactions, product information, stock monitoring, and reducing manual work. It is also a useful local reference because it comes from an Iloilo academic institution.

The proposed computerized POS and stock monitoring system aimed to improve operations and protect recorded information.

Chicano and Huervana (2009) developed a computerized restaurant sales and stock monitoring system at Central Philippine University in Iloilo City. Their study examined a manual system and identified manual billing and unmonitored stock as problems.

Local Study: Chicano and Huervana

The study also shows why historical data matters. Advanced analytics depends on reliable records. CorePOS can help create that foundation by centralizing membership, sales, and check-in information.

The study is useful when explaining the difference between basic analytics and predictive analytics. CorePOS currently focuses on organized data and basic reporting. Predictive analytics can be treated as a possible future improvement.

Sabado and colleagues (2025) presented Fit Track, a web-based gym member monitoring system with predictive analytics for Almo Fitness Gym. The system included QR attendance tracking, payment processing, and predictive analytics.

Related Study: Sabado and Colleagues

The study was conducted in Tacloban City, while CorePOS is intended for Project E Fitness. Therefore, the study should be used as supporting literature rather than as evidence about the actual client.

This is closely related to the CorePOS check-in process. Both systems recognize that attendance and membership verification can be improved using digital tools.

Villanueva, Sevilla, and Verecio (2025) developed an Olympic Fitness Gym management system with QR Code technology. Their study focused on automating sales recording, report generation, attendance monitoring, and subscription monitoring.

Related Study: Villanueva, Sevilla, and Verecio

The study is related to CorePOS because both projects aim to improve gym operations through digital management. The difference is that CorePOS is designed around its own client requirements and includes a unified POS and membership process.

The study used the Waterfall model and included user registration, login, trainer management, transaction processing, and review functions. It provides an example of a gym moving from manual processes toward a web-based system.

Muzaki, Khotimah, and Meimaharani (2025) developed a website-based management system for Angkasa Gym. Their study discussed manual member registration, training schedules, and financial transactions.

Related Study: Muzaki, Khotimah, and Meimaharani

The difference is in the scope. CorePOS focuses on Project E Fitness and combines membership, POS sales, check-in, and basic reporting. The related study is supporting evidence rather than a direct description of the CorePOS client.

The study is related to CorePOS because both projects use a web-based platform to organize gym operations that were previously handled manually.

Alboneh and Snae (2025) developed a web-based gym information system at Nahaga Gym Sabu Seba. Their study addressed manual processes involving member registration, payment verification, and membership tracking. The authors used an Agile approach and described features including online registration, package selection, digital payments, membership cards, and administrative dashboards.

Related Study: Alboneh and Snae

These categories are practical for a business system because they cover both technical performance and user experience. They also give the group a clear structure when explaining why the system is useful.

Functionality asks whether required features work. Usability asks whether users can understand and operate the system. Reliability considers consistent operation. Efficiency considers how smoothly tasks are completed. Security considers authentication, permissions, and protection of information.

The existing project evaluates CorePOS using functionality, usability, reliability, efficiency, and security.

Evaluation Criteria

The existing capstone states that testing by gym administrators and staff showed positive usability. The group should retain the actual UAT forms, ratings, comments, and other evidence in the appendices.

For CorePOS, the relevant users include gym administrators and staff. They can perform tasks such as logging in, managing a member, checking membership status, recording a sale, and viewing a report.

User Acceptance Testing, or UAT, focuses on whether intended users can use the system for the tasks it was designed to support.

User Acceptance Testing

To evaluate the operational effectiveness, performance, and user satisfaction of the **CorePOS: Unified Sales and Membership Web Platform**, User Acceptance Testing (UAT) was conducted with target stakeholders at **Project E Fitness**. The evaluator group included gym administrators/management front-desk staff and gym members for a total of 15 respondents.

The system was evaluated using a 5-point Likert Scale based on five key software quality parameters specified in the study's methodology: **Functionality, Usability, Reliability, Efficiency, and Security**.

4.2 Overview of the Developed System

Unified Sales and Membership Web Platform for a Gym is a single software platform designed with the aim of facilitating the streamlining of activities carried out at the gym's front desk, managing of membership life cycles, as well as consolidating all sales data in one web-based platform. Before the introduction of the above platform, there existed the threat of administration-related issues, data entry errors, disjointed point-of-sales data, among others at the gym. The designed platform helps address the above issues through the consolidation of gym activities such as registration, subscriptions, POS inventory and transactions, attendance management, and financial analysis into one web-based application.

- Unified Sales & Point of Sale (POS): Allows for logging of product sales (supplements, apparel, equipment) and membership plan sign-ups to facilitate financial reconciliation at one source as well as up-to-date inventory management.

- Lifecycle Management of Membership: Records the membership plan level, expiration date, renewal notifications, and member identification checks to avoid any unauthorized entries and register log entries.

- Role-Based Access Control (RBAC): Ensures data privacy and security through assigning role-based permissions to Administrators (full access and control), Staff (POS and attendance check-in), and Members (self-service).

- Analytics & Reporting Dashboard: Provides basic summaries of sales and membership information to help management monitor business performance.

The results from the study show that there is a marked decrease in operational friction in all the key functions of the gym. The shift from the use of individual physical logs and individual POS system to one web-based portal has made it possible to avoid any duplication of data entry and loss of money due to walk-ins.

Figure 4.1 illustrates the overall workflow of the developed system

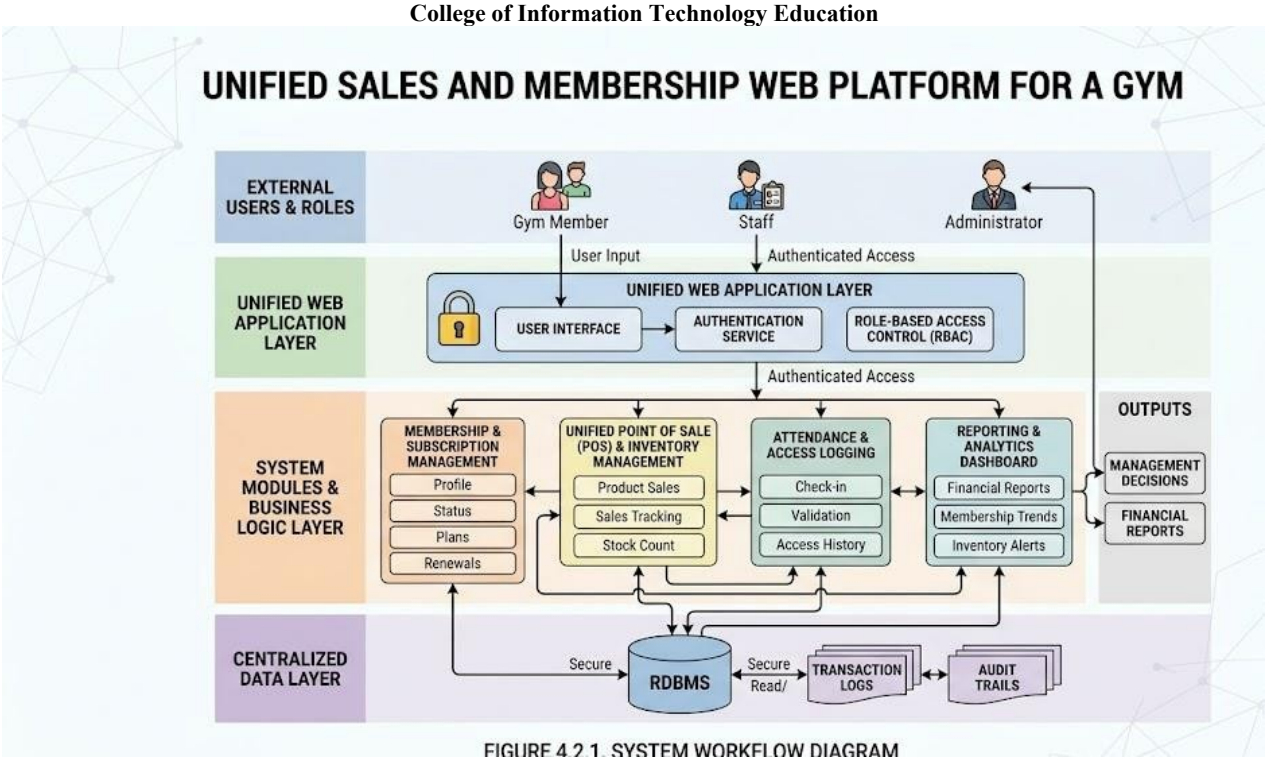


FIGURE 4.2.1. SYSTEM WORKFLOW DIAGRAM

The workflow of the Unified Sales & Membership Web Platform starts from the External User Layer, where there are three main actors – Gym Members, Staff, and Administrators who interact with the platform via a unified responsive front end interface. The access to the features of the system is controlled by the Unified Web Application Layer which verifies the users' credentials and applies the Role Based Access Control mechanism. After the authentication process is completed, the requests are forwarded to the System Modules Layer, where:

4.3 System Development Methodology

The development of the Unified Sales & Membership Web Platform for a Gym followed the Agile Software Development Methodology approach where an iterative and sprint-based process was used. The reason for using this approach instead of others is that it provides flexibility in developing software according to changing needs of operations and constant feedback from users along with the implementation of different features. Core features such as membership management, POS sales, and check-in scanner processes were developed using Agile method.

Requirements Gathering

Requirement Analysis: Interviews with front-desk employees, gym attendees, and managers helped determine pain areas within the process and establish required functions, including real-time transaction management at the POS system and membership validation.

System Planning

System Planning: In this stage, the architectural structure of the system was laid out. A roadmap for the whole project and a feature backlog were created in order to prioritize the important features and milestones of the development process.

System Design

During the coding stage, the designed modules were implemented using the chosen programming languages and frameworks. System functionalities were developed in phases and included features such as user authentication, product management, recording transactions, customer profiling, creating analytics, and generating reports.

Software Development

System Design: The coding process entailed designing the system modules through the application of chosen programming languages and frameworks. Plans were prepared for the architecture, wire frames for the user interface design and schemas for the relational database to guarantee smooth data flow.

Testing

The process of integration testing was carried out to ensure the interaction of modules works well. System testing was used to check the performance of the whole system.

The system was tested for accuracy of data and presence of bugs through unit, integration, and user acceptance testing.

Deployment

The deployment stage focused on preparing the web-based system for use after development and testing. The team checked the system configuration, database, user access, and major functions before use. Deployment and maintenance procedures were documented so that authorized gym users can operate the system and the team can address future updates or corrections. Online payment gateways, native mobile applications, and advanced analytics are outside the current scope.

Maintenance

Database maintenance includes regular backups, system updates, performance checks, and security patches.

Figure 4.2 presents the Agile Software Development Life Cycle adopted in the study.

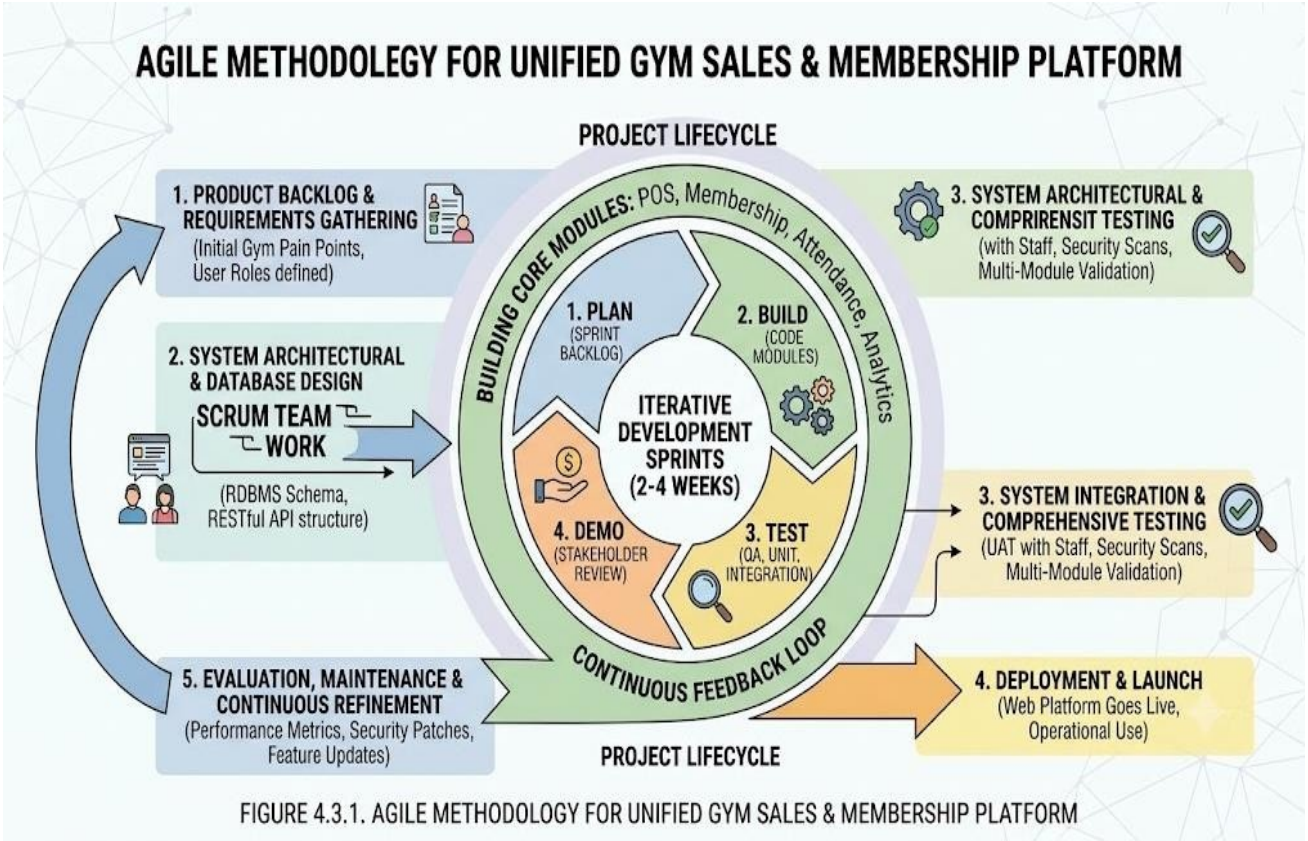


FIGURE 4.3.1. AGILE METHODOLOGY FOR UNIFIED GYM SALES & MEMBERSHIP PLATFORM

4.4 System Architecture

The technology is developed through the use of a simple three-tier architecture in order to ensure speed, security, and organization. The first tier is that of the User Interface (Front End), which consists of user-friendly web pages that allow members, employees, and administrators to have easy access to tools from any browser. The second tier involves the Application Server (Back End), which is the engine of the whole application and runs real-time business logic calculations such as sales amounts, reduction in stock, and validation of subscription activity. All of this is supported by the third tier known as the Database Layer, which is a centralized relational database that holds all information related to members, transactions, and inventory.

Figure 4.3 illustrates the architecture of the developed system.

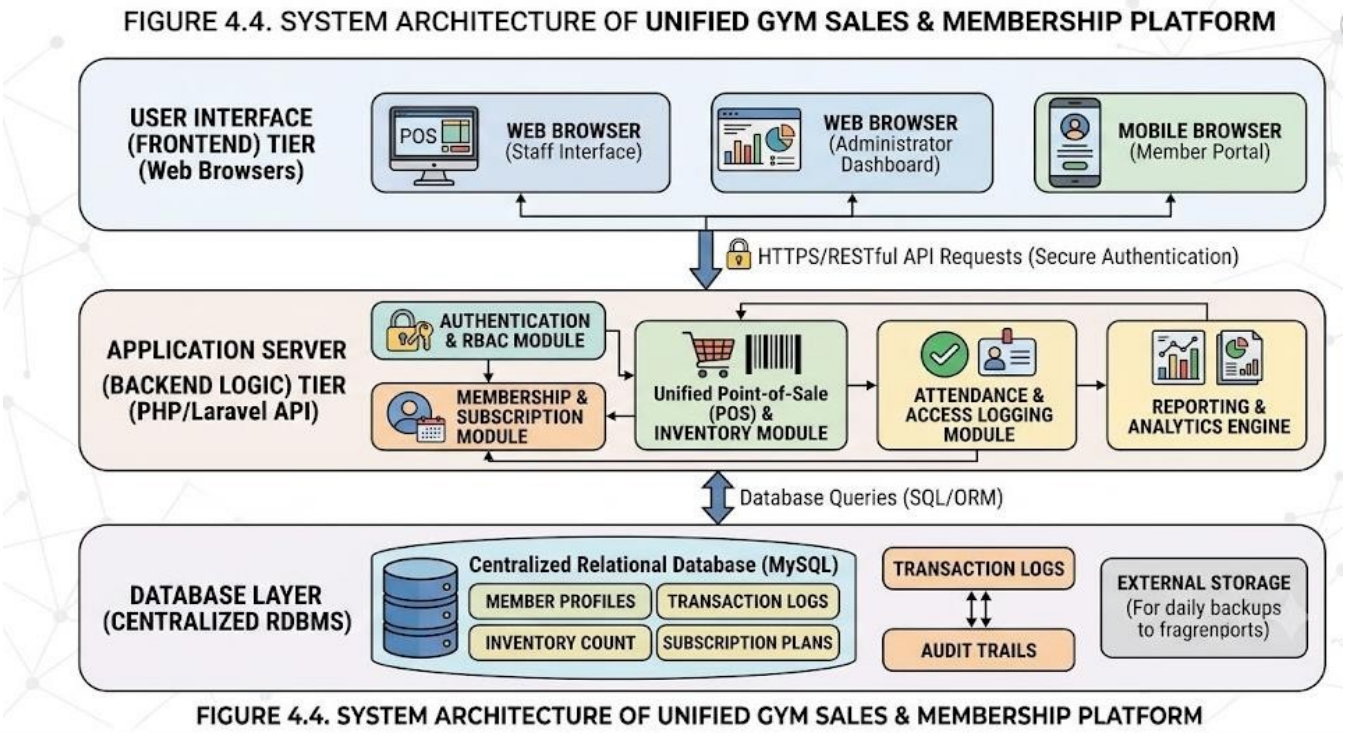


FIGURE 4.4. SYSTEM ARCHITECTURE OF UNIFIED GYM SALES & MEMBERSHIP PLATFORM

Application Layer

Data Layer

Data Layer is used to provide the basic backbone to the system and is controlled by an RDBMS (Relational Database Management System). This layer is meant to ensure that data integrity is preserved and supports fast transactions.

- Membership
- Subscriptions
- Transaction>Attendance
- Inventory/POS

4.5 Hardware and Software Requirements

To ensure optimal performance, reliability, and low-latency processing for the Unified Sales and Membership Web Platform, specific hardware specifications are required for both the hosting server environment and the workstation terminals (front-desk POS and member access terminals).

4.5.1 Hardware Requirements

Component	Minimum Specification	Recommended Specificatio
Processor	Intel Quad-Core Xeon / AMD EPYC (2.4 GHz or higher).	Intel Core i5 or higher
Memory (RAM)	8 GB minimum	8 GB or higher
Storage	128 GB NVMe SSD or higher	512 GB SSD
Monitor	(2560 x 1440)	Full HD Display
Internet Connection	50 Mbps	90 Mbps
Mobile Device	Android 11	Android 13 or higher

4.5.2 Software Requirements

Table 4.2 presents the software requirements.

Software	Purpose
Visual Studio Code	Source code editor
MySQL	Database Management
Php	Backend Development
Google Chrome	Web Browser
Git	Version Control
XAMPP	Local Development Server

4.6 Database Design

Users Table

Field	Description
Id	primary key
Username	login username
Password	user pass
Role	user access
Recovery question	recovery pass
Recovery answer	recovery pass
Created_at	date

Customers Table

Field	Description
ID	user id
Member ID	Member info
Scan time	QR Scan

Products Table

Field	Description
ID	primary key
Name	product name
Price	product price
Stock	item
Image_url	picture

Transaction Details Table

Field	Description
Id	primary key
Qr_id	user QR
Name	username
Contact	user number
Address	user address
Membership type	Membership pass
Start date	user date
Expiry date	membership end

Status	active/inactive
Created_at	date created

Analytics Table

Field	Description
Id	primary key
Sale id	user id
Product ID	product
Qty	amount
Price	amount

CHAPTER V

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Reference and Evidence Guide

User training is another part of maintenance. Staff should know how to manage a member, process a sale, perform a check-in, and generate reports. A short user guide can reduce confusion and help the business continue using the system after the developers finish the project.

The administrator should review user accounts and disable accounts that are no longer needed. The development team should also keep a record of system changes so that problems can be traced when updates are made.

After deployment, CorePOS should be maintained through database backups, account management, software updates, and monitoring.

Implementation and Maintenance

Another possible improvement is equipment usage tracking. The existing document mentions investigating IoT sensors for equipment. Such information could help management plan maintenance. These improvements would require new requirements, development, testing, and evaluation.

A future version could include attendance forecasting, membership renewal prediction, or product demand forecasting if enough historical data becomes available. A mobile application could provide members with easier account access.

The current project identifies possible future technologies such as mobile applications, machine learning, and cloud computing. These should be treated as future work rather than completed features.

Future Improvements

The system is also designed around one selected gym. Other gyms may have different membership plans, payment procedures, products, and roles. Applying CorePOS to another business would require another requirements analysis.

Another limitation is data quality. A centralized system can reduce duplicate records, but it cannot completely prevent incorrect information if users enter incorrect values. Validation and user training remain important.

The current project scope limits CorePOS to a web-based system and excludes a dedicated native mobile application and advanced analytics or AI.

Project Limitations

These qualities matter in a gym because front-desk work can be fast-paced. A system with many features is not useful if a simple check-in takes too many steps. The interface should therefore remain simple while still applying validation and access controls.

Functionality means that the required features perform their intended tasks. Usability means that users can understand and operate the system. Reliability means that the system works consistently. Efficiency concerns the time and resources needed to complete tasks. Security concerns protecting information and limiting access.

Non-functional requirements describe how the system should work. The current capstone evaluates CorePOS in terms of functionality, usability, reliability, efficiency, and security.

Non-Functional Requirements

Reporting should organize stored information into summaries. These functions are connected. A member record supports check-in. A product record supports a sale. A sale can support inventory monitoring. The resulting information can then support reports. This connection is one of the main reasons for developing a unified platform.

Membership management should allow authorized users to create and maintain member information. The sales function should record transactions for gym products or services. The check-in function should identify a member and verify membership status before recording attendance.

Functional requirements describe what CorePOS should do. The main functions in the existing project are membership management, sales recording, QR or barcode check-in, basic reports and dashboards, authentication, and role-based access control.

Functional Requirements Overview

Role-based access control is useful because not every user should be able to change every record. A staff member does not need the same permissions as an administrator. A member should not be able to change financial records. Assigning permissions by role makes access easier to control and manage.

The administrator has the broadest responsibility. This role can manage users and view business information such as sales and reports. Front-desk staff focus on daily operations, including member assistance, check-in, and sales. Members have more limited access and mainly use functions related to their own account.

CorePOS identifies three main user groups: administrators, front-desk staff, and members. Each group has a different responsibility, so each group should have different access.

User Roles

The remaining problems involve limited reporting and possible data errors. A centralized system does not remove every possible mistake, but it gives the business a structured place for validation, access control, and reporting.

The third problem is membership verification. The staff need to know whether a member is active before recording an entry. The system uses membership status and expiration information for this purpose.

The second problem is scattered membership and sales records. Separate records make it harder to compare information and prepare complete reports. Centralized tables allow related information to be retrieved more easily.

The first problem identified by the group is manual and time-consuming work. Repeated writing and checking can slow down front-desk operations, especially during busy hours. CorePOS addresses this by providing forms and search functions connected to a central database.

Problem Analysis

For a product sale, the staff selects the product, enters the quantity, checks the transaction information, and saves the sale. The transaction becomes part of the database and can support inventory and reporting. This shows why the modules are connected rather than separate applications.

For a member check-in, the system receives the member identifier and checks the membership record. If the membership is active, the check-in can be recorded. If it is expired or inactive, the status can be shown before the staff continues.

A normal CorePOS front-desk process starts when a staff member logs in using an authorized account. The staff member selects a task such as member registration, membership checking, check-in, or sales.

Business Process Mapping

CorePOS proposes a central process. An authorized user enters information, the application checks and processes the request, and the database stores the result. The stored information can then be used by the appropriate module. This gives the business one organized source of operational records.

The problem is not only the amount of paper. The bigger concern is that information may be duplicated, missed, or updated in one place but not another. This can affect membership verification, sales monitoring, and report preparation.

The current capstone describes the client's previous process as manual or divided among separate systems. Manual records can work when transactions are few, but they become harder to manage as membership and sales increase.

Existing Process Analysis

When these records are handled separately, staff may need to check several sources before completing one task. CorePOS addresses this by placing related operations in one web platform. The project is therefore not only about replacing paper records with a website. It is mainly about improving the way business information moves from one activity to another.

Membership information includes the member's details and membership validity. Sales information includes products, quantities, prices, and transaction details. Check-in information records the member's visit. Management also needs summaries of these records when reviewing business performance.

Project E Fitness is the client environment for the CorePOS project. The business provides fitness services and also handles memberships and retail products. These activities create different records that need to be organized. **Project Context**

CorePOS is treated as a practical web-based system. Its main purpose is to bring membership management, sales recording, check-in, and basic reporting into one platform. The system is connected to the needs of the selected client, so the discussion focuses on actual gym operations instead of adding unrelated features.

CONCLUSION AND RECOMMENDATIONS

5.1 Introduction

This study aimed to design and develop unified Sales & Membership Web Management Platform refers to a comprehensive web-based management platform that seeks to bring all the core functionalities of the gym into a single system to help modernize and integrate the processes. This system was developed to solve the problems associated with scattered data recording, manual attendance recording, and sales recordings.

The system helps the user and the owner to improve the business and to reduce time for the user to login and walk in the same time.

5.2 Summary of Findings

The findings of the platform evaluation and user testing include:

- All-In-One Operation: The system allows users to manage membership, conduct check-ins and conduct sales operations in one tool. There is no need for staff to open other applications or use paper logbooks.
- Automatic Membership Expiration Dates Checking: The system checks membership expiration date, so there is no possibility of allowing access of non-paying or expired members.
- Faster Check-Ins: The system supports QR-based member check-in and membership verification, which can reduce the time required for manual checking during busy periods.
- Real-Time Sales and Inventory Management: With every sale, the inventory is updated and sales are recorded in the system. The system allows for accurate tracking of revenue and prevents losses and errors due to manual sales accounting.
- Role-Based Access Control: Only the users who need to see certain information will be able to do that because of this system. Users can process sales and check-ins, but only administrators can view financial statements.
- Positive User Experience: The testing of the system by gym administrators and staff proved the usability of the system.

5.3 Conclusion

The Unified Sales and Membership Web Platform serves to showcase how the process of upgrading gyms can be achieved through the implementation of an integrated and digital system that would eliminate the problems associated with manual record keeping, disjointed sales record, and inefficient data inputting. Through combining all of these elements into one system, the purpose of the paper has been fulfilled.

A unified backend database is implemented to ensure that all activities like product purchases, membership renewals, and member check-ins take place in a safe manner in real time. Access control using roles is applied to ensure that the data remains in its proper form without compromising on usability. In the end, this system provides a practical and safe method for managing the gym that prevents revenue loss and bottlenecks at the front desk.

5.4 Recommendations

From the analysis and results of the study conducted, the following suggestions are provided to further improve and strengthen the features of the Unified Sales and Membership Web Platform for future implementation:

- Add Online Payment Options: Connect online payment methods (such as GCash, PayMaya, or credit card gateways) to the system. This will give members the convenience of renewing their subscriptions or paying for items online anytime.
- Enhance Dashboard Analytics: Improve the administrator dashboard with smart reporting features that forecast busy gym hours, track member retention rates, and automatically alert staff when product stock is running low.
- Integrate Mobile App Support: Expand the platform by adding a mobile app for gym members. This will allow members to view their membership status, check payment history, and use a digital QR code on their phones for quick front-desk check-ins.
- Automated Notification System: Integrate SMS and email notification gateways to automatically alert members regarding upcoming subscription expiration dates, payment receipts, and special gym announcements.

System Improvements

- Automated Online Payment Gateway: Integrate third-party e-wallets and card processors (e.g., GCash, PayMaya, Stripe) to allow members to pay for membership renewals or product pre-orders directly online.
- Class & Fitness Progress Tracking: Expand system features to include class scheduling (e.g., Zumba, yoga) and client workout/fitness progress logs for personal trainers.
- Automate Member Retention Tracking: Configure system reports to flag "at-risk" members who haven't checked in for 14+ days, allowing management to automatically send re-engagement promos or check-in messages.

Future Research Directions

- Predictive Analytics for Member Churn: Research machine learning models (e.g., logistic regression, decision trees) to analyze member attendance patterns, POS spending behavior, and demographic data to predict customer churn and recommend targeted retention strategies.
- Smart Attendance and Capacity Forecasting: Develop predictive time-series models to forecast daily peak gym attendance hours, enabling facility managers to optimize staff scheduling and equipment maintenance.
- Gym Equipment Usage Tracking: Investigate IoT sensors attached to gym equipment to track usage frequency and wear-and-tear, feeding real-time maintenance data back into the central database.

Implementation Strategies

- Peripheral Calibration: USB/Bluetooth QR-based check-in equipment and 80mm thermal printers are plug-and-play calibrated for instant response upon scanning member passes or checking out items.

- Member Onboarding: Quick-reference guides and digital tutorials demonstrating how to access the member web portal and check membership statuses.
- Administrator Training: Focuses on user role creation, database record overrides, system log audits, and financial reporting dashboards

5.5 Impact of the Study

There is much value in using the Unified Sales and Membership Web Platform for both the strategic and technical aspects of gym management. Gym owners will benefit greatly from this application since it will end up ensuring that no money is lost due to inefficient operations or because of an untidy record-keeping system. This is made possible by integrating sales transactions, membership renewal, and inventory information into one centralized database. With real-time sales and attendance records, gym managers can easily make important decisions regarding business matters and also cut down on licensing costs for software. On the other hand, the staff at the reception desk will benefit from the fact that most of their work will be automated through the use of barcodes and expiration checks.

The first and foremost advantage of this architecture from a consumer point of view is a contemporary, seamless user experience whereby customers can enter facilities without much ado and have complete visibility via a website that allows them to see the validity of their membership and past purchases. From an academic/technical point of view, this paper offers a strong, scalable architecture model which shows how to leverage web technology and database concepts and RBAC to transform the operations of small-to-medium enterprises. The main advantage of the proposed architecture is that it provides an easily expandable platform for integration with future technologies like mobile apps, machine learning, cloud computing, etc.

5.6 Summary

The study developed CorePOS: Unified Sales and Membership Web-Platform for Project E Fitness to address difficulties caused by manual and separated records. The systembrings membership management, sales recording, QR-based check-in, product monitoring, and basic reporting into one web-based platform. Based on the documented system functions, CorePOS provides a more organized wayof handling daily gym transactions. Member information can be maintained in one place,membership status can be checked during entry, sales can be recorded through thePOS module, and product information can be monitored through the database.The study also shows the importance of organized data for business analytics. CorePOS does not use advanced artificial intelligence within its current scope. Instead, stored sales, membership, and check-in records are used for basic summaries that can help management review business activity and support day-to-day decisions. Overall, the study indicates that a centralized web-based system can reduce repetitive manual work and make information easier to retrieve. Its usefulness still depends on correct data entry, proper user access, regular backups, and actual user evaluation. Future improvements may include online payments, mobile support, notifications, and advanced analytics if required by the client.